

AIRLINE LOYALTY PROGRAM

What the Data Tells Us — and What We Should Do About It

Consulting & Analytics Club

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Written for a smart non-technical reader

1. The Problem We Were Asked to Solve

Every airline loyalty program faces the same quiet threat: members who stop flying but never formally quit. They sit in the database, receive emails, cost money to maintain — and then one day they're gone. The marketing team knew some customers were disengaging, but had no way to tell who, how soon, or why.

We were given three specific questions to answer:

- Which customers are most likely to disengage before it's too late to act?
- Which members are genuinely valuable — and which just look that way on paper?
- What should we actually do differently for each type of customer?

"The loyalty card tier a customer holds — Star, Nova, or Aurora — turned out to be nearly useless for predicting whether they would leave. What matters is behaviour, not badge."

To answer these, we worked with the airline's full customer database — 16,737 Canadian loyalty members and over 390,000 months of flight records spanning 2017 and 2018.

2. How We Prepared the Data — and Why It Mattered

Good analysis requires clean data, and cleaning data requires judgment — not just software. Here are the three most consequential decisions we made before any modelling began.

Decision 1: Defining What 'Churned' Actually Means

The database has a cancellation field. Only 2,067 out of 16,737 members ever formally cancelled — that's just 12%. If we had relied on that field alone, we would have declared the loyalty program healthy and missed the real story.

Instead, we defined a member as churned if they formally cancelled OR if they had zero flight activity in the final six months of the observation period. This surfaced 2,686 churned customers — 16% of the base — including hundreds of silent quitters who simply stopped flying without ever clicking 'cancel'.

Key lesson: most customers don't announce when they leave. They just quietly stop showing up.

Decision 2: What to Do with Missing Salary Data

25% of members had no salary on record. A naive approach would be to remove those rows or fill in an average — both of which would be wrong. Investigation revealed that every single member with a missing salary was categorised as a College student. Their salary isn't missing: they simply don't have one yet. We set their salary to zero and kept all 4,238 of them in the analysis.

Decision 3: Removing Duplicate Records

The flight activity file contained 1,922 exact duplicate rows — records that appeared to have been entered twice. These were quietly removed. Leaving them in would have made some customers look like they flew twice as much as they actually did, distorting any engagement metric we calculated.

Data source 1	16,737 customer profiles — demographics, card tier, CLV, enrolment history
Data source 2	392,936 monthly flight records — flights, distance, points earned and redeemed
Observation window	January 2017 – December 2018 (24 months per customer)
Final analysis dataset	16,737 customers × 28 behavioural signals — zero missing values

3. The Logic Behind Our Model Choices

We built two distinct tools — one to predict individual churn risk, one to group customers into meaningful types. They answer different questions and are used differently in practice.

Tool 1: The Churn Prediction Model

We trained a machine-learning model to look at a customer's full 24-month history and produce a single number: the probability that this person will disengage. A score above 50% is flagged as High Risk.

Why this algorithm? It handles messy, real-world data well. It doesn't assume all factors matter equally. It lets us see which signals drove each prediction — which is critical for explaining results to the business, not just reporting a score.

The model was tested on customers it had never seen before. It correctly identified 95 out of every 100 customers who were going to churn — missing only 5%.

96%

Overall Accuracy

95%

Churners Correctly Caught

0.99

Prediction Confidence Score

Tool 2: The Customer Segmentation Model

Rather than treating 16,737 customers as one homogeneous group, we used a clustering algorithm to find natural groupings based purely on behaviour — how often people fly, when they last flew, how they use their points, how consistent they are.

We tested groupings from 2 to 8 segments. Three segments gave us the sharpest, most interpretable separation — fewer felt too broad, more felt arbitrary. The three groups that emerged are genuinely different from each other, and that difference shapes everything about how the airline should communicate with them.

4. The Most Interesting Things We Found

Several findings surprised us. We highlight the five that most directly change how the airline should think about retention.

1

How long ago someone last flew predicts churn better than anything else — by a wide margin.

A customer who hasn't taken a flight in the past four to six months is dramatically more likely to leave than one who flew last week — regardless of their tier, their salary, or how many points they've earned. This single signal was six times more important than the next best predictor. If the airline builds only one CRM trigger, it should be: 'member has not flown in 90 days — initiate contact.'

2

Your most frequent flyers are not necessarily loyal — and nearly one in five of them is already at high risk.

We identified a group of 813 new members who fly more than anyone else — over four flights per month on average — yet not one of them has ever redeemed a loyalty point. They treat the airline like a bus service: useful, not meaningful. 19% of this group was already flagged as high churn risk. Without intervention, the airline will likely lose them to whichever carrier offers a cheaper fare next quarter.

3

Loyalty card tier tells you almost nothing about whether a customer will leave.

We expected Aurora card holders to be the most loyal and Star card holders the most at-risk. The data said otherwise. The distribution of Star, Nova, and Aurora members was nearly identical across all three customer segments — including the most disengaged group. Targeting retention spend based on tier alone would be largely wasted effort.

4

Over half of the most disengaged customer group is already at high risk of churning — and this group makes up 25% of your base.

4,220 members — one in four customers — show all the warning signs of someone on their way out: they averaged less than one flight every four months, hadn't flown in over a year on average, and were redeeming loyalty points at an unusually high rate (often a sign someone is cashing out before leaving). 54.5% of this group scored as high churn risk. These members need immediate attention, not a monthly newsletter.

5

Nearly 70% of your customer base is stable, loyal, and flying regularly — and they rarely get any special attention.

11,704 members have been enrolled for nearly four years on average, fly consistently throughout the year, and churn at a rate of just 1%. They are the backbone of the programme. Yet because they're quiet and reliable, they rarely trigger any retention flags. Neglect is a slow-acting risk: this group deserves proactive relationship management, not just the absence of complaints.

5. Your Three Types of Customers

Every one of your 16,737 members falls into one of three behavioural profiles. Understanding which bucket a customer sits in changes everything about what you should offer them, when, and how.

⚠️ DISENGAGED AT-RISK MEMBERS	★ LOYAL CORE BACKBONE MEMBERS	✈️ HIGH-FREQUENCY NEW FLYERS
25% of members Churn rate: 57% Avg tenure: 2 years Last flew: over a year ago Burning points before leaving	70% of members Churn rate: 1% Avg tenure: 4 years Fly consistently year-round The programme's foundation	5% of members Churn rate: 11% Avg tenure: 9 months Most active flyers Never redeemed a single point

6. Three Business Recommendations

These are not data science outputs — they are business decisions that both a CFO and a CMO can act on. Each recommendation is grounded in what the data showed us, and each has a clear owner and a measurable outcome.

Recommendation 1 Build a 90-Day Inactivity Early Warning System

Who this is for: CMO (CRM & Retention team)

What to do: Set up an automated trigger that flags any member who has not taken a flight in 90 days. Send a personally framed re-engagement message within the first week of that window — before the member has mentally moved on. For members who reach 6 months of inactivity without responding, escalate to a voucher or double-miles offer. This should be a permanent CRM rule, not a one-off campaign.

Expected impact: Our model found that recency — how long ago someone last flew — is six times more predictive of churn than any other factor. Intervening at 90 days, rather than after 6–12 months, turns a high-cost win-back into a low-cost nudge. Estimated reduction in preventable churn: 20–30% among the at-risk segment.

Recommendation 2 Convert High-Frequency New Flyers into Programme Believers Before Month 6

Who this is for: CMO (Onboarding & Loyalty Engagement team)

What to do: Create a dedicated 'First Redemption' campaign targeting any member under 12 months' tenure who has accumulated points but never redeemed them. Offer a low-threshold reward — an upgrade, a lounge pass, a small voucher — that requires almost no points, purely to create the habit of redemption. The goal is not the reward itself but the emotional connection that comes from using the programme for the first time.

Expected impact: Our most active flyers have a redemption rate of exactly zero — they are accumulating points as if saving for something they never intend to claim. A member who redeems is psychologically invested in the programme; one who doesn't is just a fare-payer with a card. Getting one redemption on record before month 6 significantly reduces the probability of losing these high-frequency members to competitors.

Recommendation 3 Invest in the Loyal Core Before They Have a Reason to Leave

Who this is for: CMO + CFO (jointly — this is a budget allocation decision)

What to do: Allocate a portion of the retention budget specifically to members who have never triggered a churn flag — the 11,704 Loyal Core members who have been with the programme for four or more years. This could take the form of annual 'tenure anniversary' rewards, surprise upgrades, or priority access to new routes. The key principle: do not wait for loyalty to erode before acknowledging it.

Expected impact: It costs significantly less to maintain a loyal customer than to win back a lost one. The Loyal Core currently churns at 1% — if that rate drifted to 5%, the programme would lose roughly 580 long-tenured, high-value members per year. Preventive investment in this group is the highest-return retention spend available to the airline.

A Final Note

The work described in this report represents a shift in how the airline can think about its loyalty members — from a static list of card tiers to a living, dynamic view of who is engaged, who is drifting, and who needs a reason to stay.

The models and tools we have built are not one-time deliverables. They are designed to be refreshed as new flight data comes in, kept in sync with actual customer behaviour, and used by the marketing team to make decisions that are rooted in evidence rather than instinct.

The airline already has the data it needs to retain far more customers than it currently does. The opportunity is not to collect more information — it is to act on what the information already says.